

The models and algorithms, explained

Every algorithm, model and scoring system in the platform. Each one is explained twice: once in plain terms with a worked example, and once in the terms an engineer or a bank's risk team would use.

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How to read this. Each section opens with a green panel headed *in plain terms*. If you read only those, you will understand what every part of the platform does and why. The grey panels headed *technically* contain the same thing written for somebody who has to build it or audit it.

There are seven systems in total: the credit score, eligibility, pricing, collection, default protection, time value, and rewards. Everything else in the product is built out of these seven.

1 The credit score

IN PLAIN TERMS

The score answers one question: how much can this person be trusted to owe? It runs from 300 to 850, the same range a bank in most countries would recognise, and it decides which positions in a circle a member is allowed to claim.

It is not a tally of points that go up and down. It is recalculated from scratch every time something happens, from five things about the member's behaviour. Think of it as five separate marks out of ten, each worth a different amount, added together and scaled onto the 300 to 850 range.

The five inputs, and why each one is there

Payment record, worth about two fifths. Every contribution the member has ever made, but weighted so recent behaviour counts far more than old behaviour. A payment missed last month counts roughly twice as heavily as one missed a year ago. This is deliberate. A person who had a bad six months and then paid perfectly for a year has demonstrably changed, and a scoring system that cannot see that is punishing history rather than measuring risk.

How much is currently owed, worth about a quarter. Not how much they have borrowed in total, but how much they owe their circles right now measured against what they can afford. Somebody carrying a small obligation relative to their income is safe almost regardless of history, because they can absorb a bad month. Somebody carrying a large one is fragile even with a good history.

How long we have watched them, worth about a fifth. Circles completed and months on the platform, with completions counting much more heavily. A completed circle is worth more than a year of good months, because a completion proves the member kept paying *after* they collected the pot, which is the only moment their incentives were genuinely tested.

How many circles were joined recently, worth a tenth. This one runs backwards: joining three circles in three months lowers the score. In consumer credit this pattern is one of the most reliable early warnings there is.

Somebody suddenly reaching for several sources of money at once is usually in trouble, not doing well.

Which products have been used, worth a twentieth. Breadth of behaviour, weighted lightly because it is the weakest of the five and should never dominate.

TECHNICALLY

The score is a weighted linear model over five normalised components, each bounded to the interval zero to one, mapped onto the 300 to 850 range:

$$\text{score} = 300 + 550 \times [0.40 \cdot P + 0.25 \cdot U + 0.20 \cdot H + 0.10 \cdot C + 0.05 \cdot M]$$

P, payment record. An exponentially weighted mean of on-time outcomes, with the weight on an observation decaying by a factor of 0.95 for each month of age. A miss twelve months old therefore carries about 0.54 of the weight of one last month. Computed as the sum of decayed on-time indicators over the sum of decayed weights, so it is a true proportion rather than a raw count.

U, liability utilisation. One minus current forward liability divided by the affordability ceiling, clamped to the unit interval. Forward liability means what the member still owes across all live circles, which for a member who has collected is their remaining contributions and for one who has not is zero.

H, history. A blend, sixty per cent on completed circles capped at five, forty per cent on months observed capped at twenty four.

C, concurrency. One minus circles joined in the trailing ninety days divided by three, floored at zero.

M, product mix. Distinct product types used over four.

The visible plus and minus events are reason codes, generated by ranking each component's contribution to the gap between the member's score and the band ceiling above them. This mirrors how a bureau scorecard emits adverse action codes: the model produces the number, the reason codes explain the largest negative contributors.

A worked example

A member with: 90 per cent of payments on time once weighted for recency, owing Rs 60,000 against a Rs 200,000 ceiling, two circles completed of five, fourteen months observed, one circle joined in the last quarter, and two product types used.

$$P = 0.90 \rightarrow 0.40 \times 0.90 = 0.360$$

$$U = 1 - (60,000 \div 200,000) = 0.70 \rightarrow 0.25 \times 0.70 = 0.175$$

$$H = 0.6 \times (2 \div 5) + 0.4 \times (14 \div 24) = 0.473 \rightarrow 0.20 \times 0.473 = 0.095$$

$$C = 1 - (1 \div 3) = 0.667 \rightarrow 0.10 \times 0.667 = 0.067$$

$$M = 2 \div 4 = 0.50 \rightarrow 0.05 \times 0.50 = 0.025$$

$$\text{Total } 0.722 \rightarrow 300 + 550 \times 0.722 = 697, \text{ the Good band.}$$

Notice what would move that member most. Lifting the payment record from 0.90 to 0.95 adds eleven points. Paying down half the outstanding liability adds nineteen. Completing one more circle adds twelve. Those three

numbers are the reason codes the member sees, ranked in that order, so the advice the application gives them is the same advice the model would give itself.

The bands, and what they are for

BAND	RANGE	WHAT IT UNLOCKS
Rebuilding	Below 550	The last two positions only. May not buy a position on the marketplace
Fair	550 to 649	The second half of the order. May buy
Good	650 to 749	Any position. May buy
Excellent	750 and above	Any position, and preferential fees

Above every band sits one rule that ignores the score entirely: every brand new member is confined to the last three positions until they have completed two circles cleanly and passed a manual check. The reason is in section 5. A member the platform has never observed is not low risk or high risk, they are *unmeasured*, and the correct response to an unmeasured risk is to bound it rather than to guess at it.

Mapping to the credit bureau

IN PLAIN TERMS

Halqa's score is not the score a bank will see. A bank sees the bureau's own score, computed by the bureau from the data everybody furnishes to it. Halqa's job is to send good data, not to send its own opinion.

TASDEEQ, which is the trading name of Aequitas Information Services Limited and the first bureau licensed under the Credit Bureaus Act 2015, publishes a score running from 200 for very poor to 600 for excellent. It builds that score from six inputs: overdues, demographics, loan size, multiple borrowing, credit history length, and institution type.

Five of those six are things Halqa already captures. Overdues come from the lateness ladder, credit history length from the join date, loan size from the total commitment, multiple borrowing from the concurrency cap, and institution type from how Halqa registers as a furnisher. The sixth, demographics, the bureau derives itself from the identity number. So the alignment is close to exact, which is unusual and worth saying out loud in a partnership conversation.

TECHNICALLY

For display purposes only, a linear map exists between the two ranges:

$$\text{bureau equivalent} = 200 + (\text{score} - 300) \times 400 \div 550$$

This is presentational, not predictive. The bureau computes its own score from its own model over the full market's data, and Halqa's furnished tradelines are one input among many. Nothing in the platform should ever branch on a predicted bureau score.

The furnishing schema treats a circle position as an installment obligation. Original amount is the contribution multiplied by the number of rounds, the installment is the contribution, the term is the number of rounds, and the current balance is the forward liability. Status maps Halqa's finer internal ladder onto the bureau's standard buckets of current, one to twenty nine days, thirty to fifty nine, sixty to eighty nine, ninety plus, and written off.

The opened date is the join date, not the collection date. This is the single most consequential field in the whole integration and is discussed below.

Why the join date matters more than anything else in this section. A member who joins in January and collects in November has, by the time they collect, eleven months of perfect payment behaviour. If the tradeline opens on the join date, all eleven months are formally reported and the member has a real credit history before they have received a rupee.

If it opens on the collection date, which is what a careful accountant would choose because that is when a balance first exists, every one of those eleven months is discarded. The member gets one month of history instead of eleven.

That single field is the difference between a credit building product and a payment ledger.

2 Eligibility, and how positions are gated

IN PLAIN TERMS

Two separate gates, and a member must pass both. The first asks whether they can afford it at all. The second asks which position in the order they have earned the right to take.

The second gate is the important one, and the reason is worth understanding properly, because everything else in the risk model follows from it.

Why position is the whole game

When a member collects the pot, they receive the entire amount and still owe every contribution that remains. That remaining obligation is the only thing at risk anywhere in the arrangement. It is largest for whoever collects first, falls by one contribution with each later position, and is literally nothing for whoever collects last.

So risk is not spread evenly across a circle. It sits almost entirely in the early positions and decays to zero. A member in the last position cannot cost anybody anything, no matter who they are. A member in the first position can cost nearly the whole pot.

This changes what the platform needs to build. It does not need a system for judging whether somebody is a good person, which is hard, subjective and easy to get wrong. It needs a system that decides which member may occupy which position, which is narrow, measurable and testable.

TECHNICALLY

Each band carries a ceiling on forward liability at the moment of collection, expressed in contributions rather than in rupees so that one rule works for a circle of any size:

$$\text{earliest position} = \text{number of rounds} - \text{band ceiling}$$

BAND	CEILING, IN CONTRIBUTIONS	EARLIEST POSITION ON A TWELVE ROUND CIRCLE
Rebuilding	2	10
Fair	7	5
Good and Excellent	no ceiling	1
Any new member, whatever the band	2	10

Two ceilings, two and seven, produce the entire schedule for every circle size in the product. This is why "position five of twelve" appears in the interface without a lookup table behind it.

The gate is applied at render time, not at submit time. Positions a member cannot hold are not displayed, so the rule produces almost no visible refusals and no rejection experience.

The affordability gate

Four conditions, all of which must hold. None of them produces a refusal message; they govern what the discovery screen displays.

CONDITION	LIMIT	WHERE THE NUMBER COMES FROM
Total contributions as a share of net monthly income	33%	Halqa's own ceiling, deliberately stricter than the regulated benchmark
Contributions plus known debt service	40%	Anchored to the State Bank's cap on the debt burden ratio for consumer financing, reduced from 50 per cent in 2021
Forward exposure if every position were the earliest	4× monthly income	A stress test rather than an expected case
Concurrent circles, income unverified	1	Bounding exposure to an unobserved member
Concurrent circles, income verified	4	Verification raises the count, never the money cap
Concurrent circles as a host, no completions	2	The control against one person assembling an artificial network of circles

One rule that is easy to get wrong and expensive when you do. A good payment history unlocks more positions, more circles and less friction. It never raises the money cap. Only fresh evidence of income does that.

Lenders who let limits escalate on good behaviour alone create the defaults they later complain about, because a member's capacity to pay has not changed just because their willingness has been demonstrated. Willingness and capacity are different variables and must be gated by different evidence.

The daily product's entry test

The daily committee admits people paid daily, who by construction have no payslip and no fixed pay date. Two routes in, and a member needs only one.

Demonstrated daily earning. Two months of evidence, from any one of: a ride hailing or delivery platform account showing daily earnings history, a photographed trading licence or shop registration together with two months of daily takings recorded in the app, or two months of Halqa's own collection record showing money present on most days.

The vault alternative. Where a member cannot demonstrate daily income, they may instead lock a holding equal to the full pot for the whole sixty days. The money remains theirs, is held by a trustee in their own name, earns a return while it sits there, and is released untouched at the end if they have paid.

TECHNICALLY

The vault is a pledge with a charge in favour of the circle's members and an irrevocable redemption instruction signed at the moment of pledging. It reduces that member's expected loss contribution to zero by construction, because any shortfall settles from their own units under an instruction that predates the shortfall. Halqa never receives the proceeds; the trustee pays the round's recipient directly.

The requirement is expected to become unnecessary once automatic collection on the instant payment rail is universal, because the collection record then constitutes the daily income evidence directly. It is retained as the conservative path and as the route for members the record cannot yet vouch for.

3 Pricing

IN PLAIN TERMS

Pricing answers one question: what does Halqa have to charge so that the money it collects in fees covers the money it loses to people who do not pay?

The honest position is that nobody knows the default rate yet, because no circle has completed. Every price in the business is therefore a model with one unmeasured input, and the first hundred completed circles will measure it and reprice everything.

What a single default actually costs

Take a member who collects and then stops paying. They have already paid in some contributions and some fees. What they cost is what they still owe, less what they have already paid in fees.

$$\text{cost of one default} = \text{contribution} \times \text{rounds remaining} - \text{fee} \times \text{rounds paid}$$

A twelve month circle at Rs 20,000 a month, with a Rs 100 fee per contribution. The member collects in round four and stops.

Still owed: $\text{Rs } 20,000 \times (12 - 4) = \text{Rs } 160,000$

Fees already paid: $\text{Rs } 100 \times 4 = \text{Rs } 400$

Net cost: Rs 159,600

Now the same member collecting in round eleven instead: they owe Rs 20,000 and have paid Rs 1,100 in fees, so they cost Rs 18,900. The same person under the same rules costs eight times more depending only on when they collected. That single fact is why position gating exists.

The point at which a member becomes free

There is a position late enough that the fees a member has already paid exceed anything they could still cost. Beyond that point they are, in risk terms, free.

$$\text{break-even position} = \text{contribution} \times \text{rounds} \div (\text{contribution} + \text{fee})$$

PRODUCT	BREAK-EVEN POINT	POSITIONS THAT CARRY NO RISK
Monthly circles	Round 11.9 of 12	Only the last one
Asset circles	Round 4.0 of 12	Nine of twelve, because the item can be sold
Daily circles	Day 52 of 60	Eight of sixty

The monthly row is the uncomfortable one and should be read carefully. On a monthly circle the fee is so small relative to the pot that essentially no position is self funding. The protection has to come from somewhere other than price, which is exactly what sections 4 and 5 are about.

TECHNICALLY

Writing N for members, T for rounds, c for the contribution, φ for the fee per contribution, d for the default rate and m for the round on which defaults land on average, the platform's net position on one circle is:

$$\text{net} = \varphi N[(1-d)T + dm] - N \cdot d \cdot c(T - m)$$

Fees from every member who paid, less what was fronted for those who did not. Setting it to zero gives the tolerated default rate and the required fee:

$$\begin{aligned}\text{tolerated rate} &= \varphi T \div [(c + \varphi)(T - m)] \\ \text{required fee} &= d \cdot c \cdot (T - m) \div [T - d(T - m)]\end{aligned}$$

PRODUCT	RATE ASSUMED	BREAK-EVEN FEE	FEE FOR 3× COVER	CURRENTLY CHARGED	COVER ACHIEVED
Known monthly	0.3%	Rs 28	Rs 83	Rs 100	3.6×
Open monthly	1.0%	Rs 92	Rs 276	Rs 100	1.1×
Asset circle	1.0%	Rs 46	Rs 138	Rs 100	2.2×
Daily circle	30.0%	18% of contribution	not affordable	15%	0.8×

Two rows are thin by construction. Open monthly at Rs 100 and the daily product at 15 per cent sit at or slightly below break even on the fee line alone, and the shortfall is met by the opening auction, the marketplace commission and late fees. That is a defensible commercial choice but it should be a stated one rather than an accident.

The decision this leaves open. Rs 100 flat on every monthly circle is simple, memorable and member friendly. On known circles it gives 3.6 times cover and is comfortably right. On open circles it gives 1.1 times, which is break even with no margin.

Raising the open circle fee to Rs 275 gives roughly three times cover. The member experiences that as Rs 3,300 a year instead of Rs 1,200, against Rs 240,000 of money moved. Whether that is worth it is a judgement about how much of the first year's risk the company wants to carry on its own balance sheet rather than in its pricing.

4 Collection

IN PLAIN TERMS

The most effective decision in the entire product is not a formula. It is *when to ask*.

Halqa does not collect on the due date. It collects on the morning the member's pay arrives, which is the moment in the month when their account balance is at its highest. The due date becomes the deadline for fixing failures rather than the moment of the attempt.

This removes an entire category of default. If the money arrived and was spent before the due date, that is a failure the platform caused by asking too late. Once you ask on payday, what remains is somebody deliberately refusing, which is a completely different problem and a much smaller one.

Finding out when payday actually is

Pakistan has no open banking regime and Halqa has no bank partner, so payday has to be established from evidence the platform legitimately owns. Four routes, in ascending order of strength.

ROUTE	WHAT IT NEEDS FROM THE MEMBER	STRENGTH
Salary slip	One photograph, reduced on the device before upload	Documentary, immediate
The platform's own collection record	Nothing at all	The strongest of the four, and free
Credit alert reading	Optional consent. Three fields leave the handset: a day, a coarse amount band, a source class	Strong, narrow by design
Account title	Handled at linking	Supporting only

TECHNICALLY

The collection record route is the interesting one because it costs nothing and improves by itself. Every collection attempt is logged against a calendar day. The first success in each month marks a day on which money was provably present in the account. Taking the modal day across observed months, and requiring agreement across two consecutive months, establishes the payday with no document and no integration.

payday = mode of { first successful collection day, per month }
confidence = count of the mode ÷ months observed · verified at ≥ 2 of 2

The elegant property is that a member who declares the wrong day is corrected automatically. The retry ladder attempts collection across the following days, finds the real one, and the modal calculation converges on it. A false declaration therefore does not need to be detected and punished; it simply fails to be confirmed, which removes the fee discount and records a single score event with its reason attached, once per window so the same error is never punished twice.

Why a payment cannot happen twice

IN PLAIN TERMS

A member on a weak signal taps pay, sees nothing happen, and taps again. The handset may well have sent both requests. On most systems that is two debits and an apology.

Every instruction Halqa issues carries a unique reference naming the exact thing being attempted: this member, this circle, this round, this purpose. When the payment succeeds, the reference is stored with the record. If a second request arrives carrying a reference that is already stored, the system returns the original receipt and moves no money.

TECHNICALLY

This is idempotency, and it is applied to every write in the money path rather than only to payments. The key is deterministic and derived rather than random, so a retry from a client that has lost its state reconstructs the same key:

key = member · circle · round · purpose

ATTEMPT	KEY ALREADY STORED	BEHAVIOUR
First request	No	Charge, write settlement, persist key inside the same transaction
Duplicate request	Yes	Return the stored result. No side effects
Network retry after timeout	Yes	Identical. The retry is free
Genuine second payment	No, different purpose	Charge. The member intended it

The same discipline covers payout release, so a retried disbursement cannot pay a member twice, and it covers the reconciliation job, which queries the aggregator by key before it retries anything. A payment that quietly succeeded but whose callback was lost is therefore never charged again.

Retries are the normal condition on Pakistani mobile networks and under serverless execution timeouts. A payment system without this property is not partly correct; it is wrong on its most ordinary day.

The eleven steps, and where the money is at each one

Schedule written, notice sent, instruction fired, aggregator presents, rail settles, result returns, settlement written, receipt issued, round checked, payout released, record furnished.

The property worth noticing is that in none of those eleven steps does the money sit with Halqa. It is with the paying member, then in transit on the rail, then with the receiving member. Halqa holds an instruction and writes a record.

TECHNICALLY

Step seven is a single database transaction that writes four things or none: the debit ledger entry, the credit ledger entry, the receipt row, and the outbound message queue row. This makes three guarantees structurally rather than by convention. A receipt cannot exist without a settlement, a settlement cannot exist without a receipt, and a notification cannot be sent for a settlement that did not commit.

Amounts are integer paisa held as big integers throughout. No floating point value appears anywhere in the money path, and rates are applied as integer basis points with integer division. Binary floating point cannot represent most decimal fractions exactly, and the residue is invisible until a reconciliation fails months later. Removing the representation removes the entire class of error.

Ledger entries are written in pairs that must sum to zero. A one sided write is rejected by the database rather than discovered by an auditor.

5 The default protection model

IN PLAIN TERMS

The model does not try to make defaulting difficult for everybody. It tries to make it pointless for almost everybody, and to ensure that the few members who could genuinely cost something are the ones who have most to lose.

Ask the question from the member's side. What do I gain by walking away? For a member in the last positions, where every new member is placed, the honest answer is nothing at all: by the time they collect they have already paid in almost everything they owe. For a member in the early positions, the gain is real, but those positions are only available to people with a long clean record they would be destroying.

The seven independent layers

LAYER	WHAT IT DOES
1 Identity	Card captured at signup, home location pinned, uniqueness enforced on identity number, telephone and electronic address. Every obligation binds to a government identity
2 Position	Band and tenure decide which positions may be claimed, bounding the worst case exposure to an unobserved member
3 Schedule	Collection on payday, escalating reminders, a balance check the evening before, and an internal grace window never shown to the member
4 Price	The late fee ladder, with matched score damage at every step so money cost and record cost move together
5 Payout controls	No payout releases while a contribution in that round is unpaid; a default on a linked circle withholds it; a shortfall is withheld from the pot up to a ceiling so a turn is never emptied
6 Consequence	Platform wide feature lock, an internal flag, and a recovery case for the outstanding amount plus penalties plus a rehabilitation fee
7 Contract	The signed undertaking, renewed weekly on open circles, and the mutual guarantee executed automatically at joining, running member to member and never to Halqa

They are independent by design. A member who gets past identity still meets position. One who gets past position still meets the payout gate. No single failure exposes a circle.

How the four products differ

	KNOWN	OPEN	ASSET	DAILY
Who covers a missed contribution	The other members	Halqa	Trustee sells the item, Halqa covers any residual	Halqa
Cover compulsory	No	Yes	Yes	Yes
Income evidence	Not required	Required for the first half	Required	Two months daily, or a vault holding
Undertaking	Once at formation	Renewed weekly	Renewed weekly	Renewed weekly
Positions	Agreed by the group	Claimed, band gated	Claimed, band gated	Set once by auction, then fixed
Tradeable later	Yes	Yes	No	No
Security held	None	None	Legal title to the item	The vault holding, where required

The asymmetry a banker will spot immediately. On a known circle Halqa fronts nothing, so a default costs the platform nothing and costs the other members roughly the missing contributions divided between them. On an open circle Halqa covers it, so the members lose nothing and the platform carries the whole amount. With one flat fee on both products, the open circle member is receiving cover for free while the known circle member pays the same fee for none. That is the strongest single argument for splitting the two prices, and it is a fairness argument rather than a revenue one.

The cover facility, and the four properties that bound it

An operator in the United States guaranteed every member's default itself, without pricing or capping the guarantee. It ran nine years and closed. Four properties prevent a repeat.

Priced. The fee is set against the default rate the product is expected to carry. If the rate proves higher, the price changes rather than the promise.

Capped at four levels. Per circle, per host, per member, and in aggregate against the platform's own capital. No single circle, organiser or bad month can consume the facility.

An advance, not a payment. Halqa steps into the circle's claim against the member who failed. The obligation changes hands rather than disappearing, and it is pursued.

Reinsured above the cap. Beyond the aggregate ceiling the exposure passes to a takaful operator, so an abnormal year does not fall on the company.

6 Time value, and why position is worth money

IN PLAIN TERMS

Every member pays the same total and receives the same total, so on paper nobody gains anything. That is true in rupees and false in economics.

Receiving a lump sum in month one is worth more than receiving the same sum in month twelve, because you have the use of the money for eleven extra months. Every member already knows this instinctively, which is exactly why they compete for early turns and why informal committees have always had quiet arguments about who goes first.

The question for a formal platform is not whether that value exists. It is whether the arrangement prices it openly or pretends it does not.

What the spectrum actually looks like

Take a twelve month circle at Rs 20,000 a month, and discount at twelve per cent a year, which is roughly what money costs an ordinary Pakistani household.

POSITION	PRESENT VALUE OF THE POT RECEIVED	PRESENT VALUE OF EVERYTHING PAID IN	NET VALUE OF THAT POSITION
1	Rs 237,745	Rs 225,829	plus Rs 11,916
4	Rs 231,077	Rs 225,829	plus Rs 5,248
6	Rs 226,769	Rs 225,829	plus Rs 940
7	Rs 224,637	Rs 225,829	minus Rs 1,192
12	Rs 214,286	Rs 225,829	minus Rs 11,543

The spread between first and last is Rs 23,459, which is 9.8 per cent of the pot. That value is not created by Halqa and it is not destroyed. It moves from the late members to the early ones.

TECHNICALLY

Writing P for the pot, c for the contribution, T for the term, k for the position and i for the monthly discount rate:

$$\text{value of position } k = P \div (1 + i)^k - c \cdot [1 - (1+i)^{-T}] \div i$$

The second term is an ordinary annuity present value and does not depend on k , so the whole spectrum is driven by the first term alone. Setting the expression to zero and solving for k gives the position at which the arrangement is economically neutral:

$$\text{neutral position} \approx 6.4 \text{ of } 12, \text{ at twelve per cent a year}$$

The neutral position moves with the discount rate. At two per cent a year the order is nearly flat and position barely matters. At thirty per cent, which is close to what informal credit costs in this market, the first position is worth over Rs 25,000 and the last costs about the same. So the same circle is a very different proposition to two members with different costs of money.

A finding worth acting on. The early position fee currently tapers to zero at the *last* position. It should taper to zero at the *neutral* position, around 6.4 of 12.

At its steepest the fee captures about Rs 4,200 of the Rs 11,916 the first position is worth, which is roughly a third. The correction is one line of logic: taper to zero at the neutral position, and credit the positions beyond it out of what the early positions paid.

The circle then settles its own time value internally, and the arrangement becomes provably fair rather than conventionally fair. That is a much stronger thing to be able to say to a regulator.

Why the marketplace exists

Members do not value their positions identically, because they do not face the same cost of money. A member with a cash need this month and a member saving for a wedding next year sit on opposite sides of this spectrum. Letting them trade makes both better off, and Halqa takes a commission on the premium rather than a position in the trade.

The daily product is the exception. Its positions are set once by auction and are never tradeable afterwards, because a sixty day cycle with several collections a day does not have enough time for a secondary market to clear honestly.

7 The reward system

IN PLAIN TERMS

On time payment is the single input the whole business rests on. Expected loss, the price, the score and the value of the data asset all move with it. So the reward system is attached to that one behaviour and to nothing else.

The clever part is who pays for it. The discount at every tier is the retailer's own promotional discount, offered to an audience that has been qualified by a year or more of demonstrated payment behaviour. For the retailer that is customer acquisition spend reaching a segment they cannot otherwise identify. Halqa earns a commission on the resulting order. So the reward is the retailer's discount, the revenue is the retailer's commission, and the system funds itself.

The design rules, and why each exists

RULE	WHY
Points and score are strictly separate	Points buy discounts. Score gates positions. Points never buy score and score is never sold. Blur this and the score stops measuring risk
Paying for another member earns points but no streak and no score	A wealthier friend cannot buy anybody a credit record. This is the single most important exclusion in the system
The score gain per cycle is capped	Otherwise a member could farm score by joining many tiny circles
A missed contribution resets the streak but confiscates nothing already earned	The punishment is the fee ladder and the score. Clawing back rewards on top is punitive and teaches nothing
Rewards release only against contributions that have actually settled	A reward against an unsettled payment is a reward for an assertion
Never redeemable for cash	Cash redeemable points are a stored value instrument, which is regulated activity. This one rule keeps the whole scheme outside that perimeter

TECHNICALLY

The ladder is a step function on consecutive clean rounds, with a multiplier on points that compounds to a ceiling. Tiers sit at three, six, twelve, eighteen, twenty four and thirty six rounds, plus a completion award. Each tier offers either a partner discount capped in rupees or a Halqa charge waived, and the member chooses.

Rollout is four staged phases, ordered by counterparty dependency: charges waived, which needs no counterparty and ships immediately; affiliate registration, which is a form rather than a negotiation; direct merchant agreements with exclusive codes and in-app placement; and direct payment, where a goal circle's pot is paid to a retailer for a named item and the retailer pays a commission on the value brokered.

Halqa receives order attribution only, never basket contents. Purchase behaviour is not sold.

8 The security model

IN PLAIN TERMS

Holding no member money removes the biggest prize an attacker could go after, but not the others. Six attacks are worth defending against, and the useful way to rank them is by what they would actually be worth to somebody attempting them, not by how alarming they sound.

ATTACK	WHAT STOPS IT
1 An impersonated exit or hardship claim	Facial verification on every exit request, the application PIN, and a manual review sample
2 Account takeover with a stolen credential	Rotating refresh tokens, reuse of an old token burning the entire family, a facial check on any new device, PIN held only in memory
3 A synthetic or borrowed identity at signup	Card capture, account title match, national database verification and biometrics, uniqueness on three separate identifiers
4 One person assembling an artificial circle	Host concurrency ceilings, the host link confirmed from both ends, new members confined to the last positions
5 A double charge or double payout through a retry	The idempotency key on every write, and reconciliation by key before any retry
6 Data exfiltration	Parameterised queries throughout, no raw biometrics or card numbers stored at all, append only audit log, encryption in transit and at rest

TECHNICALLY

Passwords are hashed with bcrypt at cost twelve, and secret strength is asserted at boot so the service refuses to start with weak secrets rather than starting insecurely. Sessions use short lived access tokens with rotating refresh tokens; presenting a previously rotated refresh token is treated as compromise and burns the entire token family.

Rate limiting locks an account for fifteen minutes after five failed attempts and blocks globally at thirty attempts in fifteen minutes. Login timing is held constant so accounts cannot be enumerated by measuring response times.

Every database query is parameterised through a typed data layer, so constructing a query by string concatenation is structurally unavailable rather than merely discouraged. An append only audit log and a separate security event table record every consent, signature and administrative action with an actor and a timestamp.

Posture is assessed internally at B plus. The gap to A is an external penetration test and a formal secure code review, both scheduled before public launch and named here rather than omitted.

What the platform refuses to collect, and why that is a security decision

Roughly four hundred lending applications were removed from Pakistani app stores for the permissions they demanded rather than for the money they lent. Every item below would improve Halqa's underwriting. Every one is refused.

REFUSED	WHY
Contact list access	The practice most associated with the removals, and the one the regulator named
Photo gallery access	No proportionate purpose, impossible to justify to a store reviewer
Message reading	Reads every private message a person has, to obtain a handful of numbers
Accessibility service screen scraping	Live bank balances with no partnership needed. Rejected early. It is the delisting trap
Background location	One precise pin at signup, with consent, is sufficient and defensible
Full card numbers and security codes	Never persisted. Brand, last four, expiry and holder only
Raw biometric templates and facial images	Only the result and a reference are kept. The template stays with the national authority

The permissions a lender demands are the fastest signal a regulator has for how it intends to collect. Halqa asks for less than a food delivery application does, and that is a deliberate positioning decision as much as a technical one.

9 What has not been measured yet

Every price in this document depends on one number nobody has: the rate at which members actually default. The figures used throughout are modelled from the academic literature on Pakistani committees, from the published experience of operators in comparable markets, and from the design points of each product.

The first hundred completed circles will measure it. At that point every fee, every band ceiling and every cover cap is recomputed against evidence rather than assumption.

Saying that plainly is a stronger position than presenting the figures as settled, because it describes the pricing as a model with one named input rather than as a guess. Anybody serious will ask the question, and the answer should be ready before they do.